$$S + O_2 \longrightarrow SO_2$$

$$4FeS_2 + 11O_2 \longrightarrow 2Fe_2O_3 + 8SO_2$$

ii. Sulphur dioxide formed is oxidised to sulphur trioxide by air in the presence of a catalyst such as V_2O_5 or platinised asbestos.

iii. The sulphur trioxide is absorbed in concentrated sulphuric acid and produces oleum ($H_2S_2O_7$). The oleum is converted into sulphuric acid by diluting it with water.

$$SO_3 + H_2SO_4 \longrightarrow H_2S_2O_7 \xrightarrow{H_2O} 2H_2SO_4$$

To maximise the yield the plant is operated at 2 bar pressure and 720 K. The sulphuric acid obtained in this process is over 96 % pure.

Physical properties:

Pure sulphuric acid is a colourless, viscous liquid (Density: 1.84 g/mL at 298 K). High boiling point and viscosity of sulphuric acid is due to the association of molecules together through hydrogen bonding.

The acid freezes at 283.4 K and boils at 590 K. It is highly soluble in water and has strong affinity towards water and hence it can be used as a dehydrating agent. When dissolved in water, it forms mono ($H_2SO_4.H_2O$) and dihydrates ($H_2SO_4.2H_2O$) and the reaction is exothermic.

The dehydrating property can also be illustrated by its reaction with organic compounds such as sugar, oxalic acid and formic acid.

$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11}} + H_2SO_4 \longrightarrow 12C + H_2SO_4.11H_2O$$

$$\underset{\text{Formic acid}}{HCOOH} + H_2SO_4 \longrightarrow CO + H_2SO_4.H_2O$$

$$\underset{\text{Oxalic acid}}{(COOH)_2} + H_2SO_4 \longrightarrow CO + CO_2 + H_2SO_4.H_2O$$

Chemical Properties:

Sulphuric acid is highly reactive. It can act as strong acid and an oxidising agent.

Decomposition: Sulphuric acid is stable, however, it decomposes at high temperatures to sulphur trioxide.

$$H_2SO_4 \longrightarrow H_2O + SO_3$$

Acidic nature: It is a strong dibasic acid. Hence it forms two types of salts namely sulphates and bisulphates.

$$H_2SO_4 + NaOH \longrightarrow \underset{\text{sodium bisulphate}}{NaHSO_4} + H_2O$$

$$H_2SO_4 + 2NaOH \longrightarrow \underset{\text{sodium sulphate}}{Na_2SO_4} + 2H_2O$$

$$H_2SO_4 + 2NH_3 \longrightarrow \underset{\text{Ammonium sulphate}}{(NH_4)_2SO_4}$$

Oxidising property: Sulphuric acid is an oxidising agent as it produces nascent oxygen as shown below.

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